

MediVision

CNN-Based Multi-Domain Medical Image Screening

We built a system that looks at a photo of your skin, eye, or teeth and tells you what might be going on — automatically.

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Why we built MediVision

The problem

You notice a weird spot on your skin, or your eye looks red, or a tooth hurts. What do you do?

Most people wait. Specialists are hard to get, and by the time you see someone, things may have gotten worse.

Existing AI tools handle only one area. None work across domains.

Our approach

One upload, three domains. The system figures out the domain automatically and routes to the right classifier.

24 conditions across skin (11), eye (6), and teeth (7), with Grad-CAM so users can see why the model made its call.

Like a GP triage — not replacing doctors, just helping decide if you should see one.

24

conditions detected

3

clinical domains

35,884

training images

85.9%

end-to-end accuracy

Three questions we had to answer

01

Can CNNs actually work here?

Skin lesions, eye conditions, dental issues — can deep learning reach useful accuracy on public data?

Yes — transfer learning from ImageNet gets us there. ResNet50 hits 84-96% per domain.

02

Can one model route all three?

Skin photos look nothing like dental X-rays. Can a single classifier reliably sort them?

A dedicated domain router gets 99.45% routing accuracy on 5,648 test images.

03

How do we keep users safe?

If the model is wrong and someone acts on it, that's a real problem.

Grad-CAM shows what the model sees, calibrated abstention flags low confidence, and a disclaimer gate blocks everything until acknowledged.

The data we worked with

7 public sources on Kaggle & Roboflow, cleaned aggressively.

Skin

11 classes | ~18.5K images

Melanoma, BCC, Eczema, Psoriasis, Nevus,
Tinea...

Eye

6 classes | ~1.9K images

Cataract, Conjunctivitis, Glaucoma, Uveitis,
Dry Eye...

Teeth

7 classes | ~11K images

Caries, Gingivitis, Calculus, Hypodontia,
Ulcer...

What we did to clean it up

PIL magic-byte check

Full pixel decode on every file —
caught mislabelled formats

Perceptual hash dedup

16×16 grayscale hashing removed
5,855 duplicates

Stratified splitting

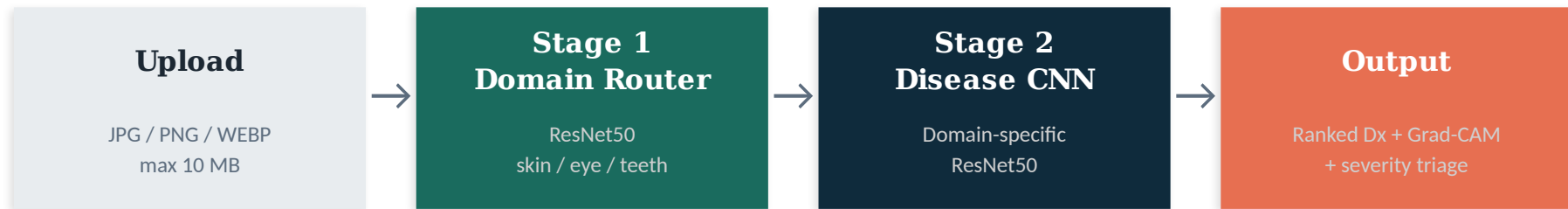
70/15/15 with leakage checks across
all splits

Class rebalancing

Capped Nevus, upsampled
minorities, focal loss, NST augment

How the pipeline works

Two stages — first figure out what kind of image, then diagnose it.



Why not just one 24-class model?

Each domain gets its own settings

Eye uses 384×384 inputs, skin uses focal loss, teeth uses standard CE.

No cross-domain interference

Tuning the skin loss doesn't accidentally hurt teeth performance.

Easy to swap parts out

Retrain just one model without touching the others.

Why ResNet50 over VGG16

We trained both on all 4 models (8 total) and compared head-to-head.

Metric	VGG16	ResNet50
End-to-end accuracy	84.03%	85.91%
Skin (11-class)	82.18%	83.94%
Eye (6-class)	93.75%	93.99%
Teeth (7-class)	94.95%	95.71%
Domain routing	98.96%	99.56%
Skin Macro F1	0.753	0.775
Feature depth	7×7×512	7×7×2048

The bottom line

ResNet50 wins every metric.

Biggest gap on skin — the hardest task — where higher macro F1 means better recall on rare malignant classes.

Skip connections, 4× richer feature maps, and better confidence calibration all favour the deeper network.

Also the industry standard for medical imaging benchmarks.

How we trained each model

Domain Router

224×224 | Single-phase | CE + label smoothing

Full backbone unfrozen from epoch 0, LR 1e-4, patience 10. Big dataset, clear visual differences.

Skin Classifier

224×224 | Single-phase | Focal loss ($\gamma=2$)

Focal loss focuses on hard/rare examples. Minorities upsampled to 2K; Nevus capped at 3.5K.

Eye Classifier

384×384 | Two-phase TL | CE + label smoothing

Smallest dataset. Phase 1: frozen backbone 15 epochs. Phase 2: unfreeze top 50 layers at 1e-5. NST augmentation.

Teeth Classifier

224×224 | Single-phase | CE + label smoothing

Balanced class weights via sklearn. Generic augmentation (flips, rotation, zoom, contrast).

Common: Adam, Dropout 0.4, L2 1e-4, label smoothing 0.1, LR warm-up 3 epochs, balanced class weights, A100 GPU

What the user sees

Flask + plain HTML/CSS/JS. No framework. All 4 models loaded at startup.

Landing Page

- Disclaimer modal blocks everything on first visit
- Three domain cards show what we detect
- Amber medical-advice banner throughout
- Consent stored in sessionStorage

Upload Screen

- Drag-and-drop for JPG/PNG/WEBP (10 MB max)
- Auto-detection — no manual domain pick
- Optional symptoms text box
- Client-side validation before hitting server

Results Screen

- Domain + confidence pills (red if low conf)
- Top-k bar chart for all domain classes
- Grad-CAM heat-map with activation legend
- Severity: Urgent / Moderate / Mild triage

Privacy: images processed in memory via BytesIO — never written to disk.

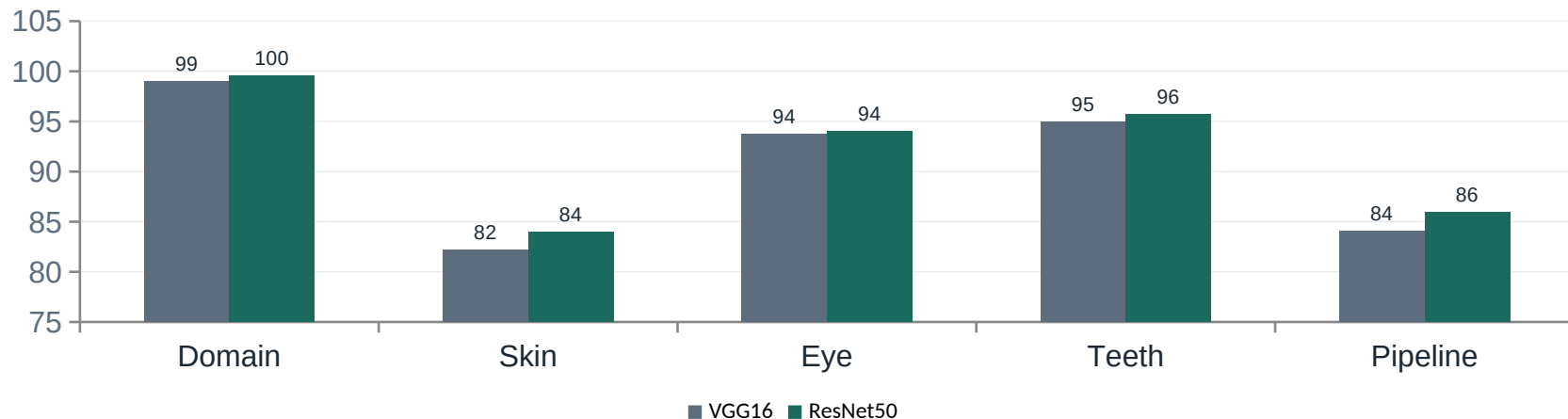
Results

85.91%

End-to-end pipeline accuracy
5,648-image stress test

Domain Routing	99.45%	Only 3 misroutes in 675 test images
Skin (11 classes)	83.94%	Melanoma F1: 0.97, BCC F1: 0.92
Eye (6 classes)	93.99%	Top-3 accuracy: 99.76%
Teeth (7 classes)	95.71%	Top-3 accuracy: 99.45%

All with bootstrap 95% CIs (1,000 resamples)



Making it safe to use

A wrong medical prediction can be dangerous. Here's how we handle that.

Grad-CAM heat-maps

Every prediction shows which pixels the model focused on. If it's looking at the wrong region, you can tell immediately. Uses last conv layer of each ResNet50.

Calibrated abstention

If top probability falls below threshold (0.80 skin/teeth, 0.60 eye), result is flagged as low-confidence. Confidence pill turns red with a warning banner.

Disclaimer gate

Full-page modal blocks all interaction until user acknowledges this isn't medical advice. Persistent amber banner on every page. Results screen recommends seeing a doctor.

Severity triage:

URGENT

MODERATE

MILD / NORMAL

What we'd improve next

Known limitations

Caries recall is only 0.53

Precision 1.00 but we miss half of actual cases. Only 36 test samples.

Skin is tough at ~84%

Eczema and psoriasis look similar even to dermatologists.

No external validation

All metrics from held-out splits. Real clinical performance could differ.

Routing could be fragile

May partly rely on lighting cues. Out-of-distribution photos might route wrong.

If we had more time

1. External clinical validation on independent datasets
2. Try EfficientNet-V2, ConvNeXt, maybe ViT
3. More caries data with consistent intraoral photos
4. Enable test-time augmentation in production
5. Shared-backbone multi-task model (~70% less GPU)

Live Demo

Let's see MediVision in action

1

Upload a skin, eye, or dental
image

2

Watch Stage 1 auto-detect the
domain

3

See ranked predictions and
confidence

4

Inspect Grad-CAM and
severity triage

Thank You

Questions & Oral Defense

CNN-Based Multi-Domain Medical Image Screening | Project 85 | 42028 Deep Learning & CNNs



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